

Program 6

Objective: Deploy an automated build pipeline by building a Docker image and pushing it to Docker Hub and GitHub Container Registry using Personal Access Tokens (PAT).

Project Structure

```
program-6/  
|  
├─ Dockerfile  
├─ package.json  
├─ package-lock.json  
└─ node_modules/
```

Steps Involved

1. Create Personal Access Token (PAT)

- Create a GitHub repository named **program-6**
- Generate PAT with scopes **read:packages** and **write:packages**

2. Create Project Directory

```
1rv24mc078_prateek_kumar@arch mkdir program-6  
1rv24mc078_prateek_kumar@arch cd program-6
```

3. Build Docker Image

```
1rv24mc078_prateek_kumar@arch ~/program-6 docker build -t program6 .
```

4. Tag Image and Login to Docker Hub

```
1rv24mc078 prateek kumar@arch docker tag program6
```

```
000prtk000/program6:latest  
1rv24mc078_prateek_kumar@arch docker login -u 000prtk000
```

5. Push Image to Docker Hub

```
1rv24mc078_prateek_kumar@arch docker push 000prtk000/program6:latest
```

6. Tag Image and Login to GitHub Container Registry

```
1rv24mc078_prateek_kumar@arch docker tag program6  
ghcr.io/000prtk000/program6:latest  
1rv24mc078_prateek_kumar@arch docker login ghcr.io -u 000prtk000
```

7. Push Image to GitHub Container Registry

```
1rv24mc078_prateek_kumar@arch docker push  
ghcr.io/000prtk000/program6:latest
```

8. Push Source Code to GitHub

```
1rv24mc078_prateek_kumar@arch git init  
1rv24mc078_prateek_kumar@arch git remote add origin  
https://github.com/000prtk000/program-6.git  
1rv24mc078_prateek_kumar@arch git add .  
1rv24mc078_prateek_kumar@arch git commit -m "Initial commit"  
1rv24mc078_prateek_kumar@arch git push -u origin main
```

9. Verification

- Verify image in Docker Hub: **000prtk000/program6**
- Verify image under GitHub → Packages
- Verify source code in GitHub repository

Output

Docker Hub & GitHub Container Registry Output:

